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Higher spin dynamics in the presence of anisotropy

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1 Introduction

While spin-$\frac{1}{2}$ particles realize qubits, recent studies suggest that higher dimensional spin particles can be useful for various quantum information applications, such as in fault tolerant quantum computation [1] and gate efficiency[2]. Such particles are called "qudits".
In order to manipulate and use these qudits, it is useful to be able to understand and control their spin dynamics. In this project, the problem of these spin dynamics was addressed for spin 1 and spin $\frac{3}{2}$ particles in the presence of a magnetic field and anisotropy, a common case for a spin sitting in a crystal as an example.

In the first part of this report, we will do an analytical study of the spin evolution for three cases : spin $\frac{1}{2}$ (which will act as a test), spin 1 and spin $\frac{3}{2}$. We will then use a Python simulation to obtain the time evolution of the spin and study those results.

2 Theoretical analysis of the Hamiltonian

In the rest of this project, the Hamiltonian will be described as the following:

$$H = -\gamma_g \boldsymbol{B} \cdot \boldsymbol{S} + K_\perp S_x^2 + K_\parallel S_z^2. \tag{1}$$

In this equation, several terms appear:

- γ_g designates the gyromagnetic ratio (i.e. the magnetic moment of the angular momentum of the particle). We chose$\gamma_g = 1$ here for more simplicity.
- $\boldsymbol{B}$ is a time-independent magnetic field. It has the form $\boldsymbol{B} = (B_x, B_y, B_z)$. In our case, $|\boldsymbol{B}| = 6$ Tesla.
- $\boldsymbol{S}$ represents the spin operator such that $\boldsymbol{S} = (S_x, S_y, S_z)$,
- $K_\perp$ and $K_\parallel$ are the anisotropy parameters. It is important to note that they are relevant only for $s \geq 1$ and become trivial for the case $s = \frac{1}{2}$. The values of $K_\perp$ and $K_\parallel$ considered here are $\frac{K_\perp}{\gamma_g B} = \frac{\pi}{10}$ and $\frac{K_\parallel}{\gamma_g B} = -0.1$

Our aim for this project is to solve the Schrodinger equation for the density operator using natural units ($\hbar = 1$) and obtain the time evolution of the spin operator $\boldsymbol{S}$.
For our case however, it is easier to use the Liouville-von Neumann equation instead of the Schrodinger equation:

$$i\dot{\rho}_t = [H, \rho_t]. \tag{2}$$

In this equation, $\rho(t)$ designates the density matrix and is defined as $\rho(t) = |\Psi(t)\rangle \langle\Psi(t)|$. The formal solution of the Liouville-von Neumann equation is:

$$\rho(t) = e^{-iHt}\rho(0)e^{iHt}. \tag{3}$$

Using the density matrix formalism, the time evolution for the spin operator is defined as:

$$\langle \boldsymbol{S} \rangle = Tr(\rho_t \boldsymbol{S}). \tag{4}$$

In this project, we will only consider pure states, therefore the density matrix will be one of a pure state. In the following subsections, we will see how to solve the Hamiltonian for $s = \frac{1}{2}$, $s = 1$ and $s = \frac{3}{2}$.

2.1 Spin $\frac{1}{2}$

We will start by studying the case of spin $\frac{1}{2}$, as it is an easy case that will act as a test later on to determine if our program is working as it should. For this spin, the two possible initial states will be $|\Psi_1\rangle = |\frac{1}{2}\,\frac{1}{2}\rangle$ and $|\Psi_2\rangle = |\frac{1}{2}\,\frac{-1}{2}\rangle$. For this case, the anisotropy terms are neglected, therefore the Hamiltonian is defined as:

$$H = -\gamma_g \boldsymbol{B} \cdot \boldsymbol{S}. \tag{5}$$

Here, the spin operator is defined by the Pauli matrices:

$$S_x = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad S_y = \frac{1}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad S_z = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{6}$$

Therefore if we consider a constant magnetic field, our Hamiltonian will be written as:

$$H = \frac{-\gamma_g}{2}\begin{pmatrix} B_z & B_x - iB_y \\ B_x + iB_y & -B_z \end{pmatrix}. \tag{7}$$

Here we will only consider a magnetic field which is only along the z-axis. Therefore we will have:

$$H = \frac{-\gamma_g}{2}\begin{pmatrix} B_z & 0 \\ 0 & -B_z \end{pmatrix}. \tag{8}$$

We chose our initial state here to be in the x-direction. The eigenvalues of the Hamiltonian are then found to be $-\frac{\gamma_g}{2}B_z$ and $+\frac{\gamma_g}{2}B_z$. Therefore, our density matrix will have the form:

$$\rho_t = \frac{1}{2}\begin{pmatrix} 1 & e^{-i\gamma_g B_z t} \\ e^{i\gamma_g B_z t} & 1 \end{pmatrix}. \tag{9}$$

After calculating the density matrix, we can find the average values of the spin operator:

$$\langle S_x \rangle = Tr(\rho_t S_x) = \frac{1}{2}\cos(\gamma_g B_z t), \tag{10}$$

$$\langle S_y \rangle = Tr(\rho_t S_y) = \frac{1}{2}\sin(\gamma_g B_z t), \tag{11}$$

$$\langle S_z \rangle = Tr(\rho_t S_z) = 0. \tag{12}$$

We can see that in the case of $m = \frac{-1}{2}$ the expectation value of the spin will oscillate and $\langle S_z \rangle$ will be equal to 0, which was expected. If the initial state was with $m = \frac{1}{2}$, the spin eigenstate wouldn't have evolved.

2.2 Spin 1

In the case of spin-1, the three possible initial state are $|\Psi_1\rangle = |1-1\rangle$,$|\Psi_2\rangle = |10\rangle$ and $|\Psi_3\rangle = |11\rangle$.
For this case, we must add the anisotropy terms and thus our Hamiltonian becomes more complicated. But first of all we must be able to express the spin operator with a 3×3 matrix system. This can be done using the Gell-Man matrices, a set of traceless Hermitian matrices which are linearly independent and are generators of the SU(3) group. Thus they can be used to rewrite the Pauli matrices in higher dimensions. They constitute the standard basis when working in a 3×3 hermitian space. All of the Gell-Mann matrices can be found in Appendix A. More information about Bloch vectors in higher dimensions can be found in [3].
Therefore the three components of the spin operator can be written as:

$$S_x = \frac{1}{\sqrt{2}}(\lambda_1 + \lambda_6) = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \tag{13}$$

$$S_y = \frac{1}{\sqrt{2}}(\lambda_2 + \lambda_7) = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \tag{14}$$

$$S_z = \frac{1}{2}(\sqrt{3}\cdot\lambda_8 + \lambda_3) = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{15}$$

We can write our Hamiltonian under matrix form using those matrices and we obtain:

$$\boldsymbol{B}\cdot\boldsymbol{S}=\frac{1}{\sqrt{2}}\begin{pmatrix} B_z & B_x-iB_y & 0 \\ B_x+iB_y & 0 & B_x-iB_y \\ 0 & B_x+iB_y & -B_z \end{pmatrix}. \tag{16}$$

We define $B=\sqrt{B_x^2+B_y^2+B_z^2}$ and we rewrite the previous equation using spherical coordinates where θ is the polar angle and φ is the azimuthal angle. We obtain:

$$\boldsymbol{B}\cdot\boldsymbol{S}=\frac{1}{\sqrt{2}}B\begin{pmatrix} \cos(\theta) & \sin(\theta)e^{-i\varphi} & 0 \\ \sin(\theta)e^{i\varphi} & 0 & \sin(\theta)e^{-i\varphi} \\ 0 & \sin(\theta)e^{i\varphi} & -\cos(\theta) \end{pmatrix}. \tag{17}$$

The set of spherical coordinates makes the form of the Hamiltonian easier but will also allow us to visualize different cases of magnetic field simply by changing the values of those angles.
The complete matrix form of the Hamiltonian can be written as:

$$H=\begin{pmatrix} \frac{-B}{\sqrt{2}}\cos(\theta)+\frac{1}{2}K_\perp+K_\parallel & \frac{-1}{\sqrt{2}}B\sin(\theta)e^{-i\varphi} & \frac{1}{2}K_\perp \\ \frac{-1}{\sqrt{2}}B\sin(\theta)e^{i\varphi} & K_\perp & \frac{-1}{\sqrt{2}}B\sin(\theta)e^{-i\varphi} \\ \frac{1}{2}K_\perp & \frac{-1}{\sqrt{2}}B\sin(\theta)e^{i\varphi} & \frac{B}{\sqrt{2}}\cos(\theta)+\frac{1}{2}K_\perp+K_\parallel \end{pmatrix}. \tag{18}$$

We can see that this matrix will be harder to diagonalize by hand due to all the different parameters. Therefore, its diagonalization and the finding of the average values of the spin operator will be obtained using a Python program.

2.3 Spin $\frac{3}{2}$

In the case of spin $\frac{3}{2}$, the four possible initial states are $|\Psi_1\rangle=|\frac{3}{2}\frac{-3}{2}\rangle$, $|\Psi_2\rangle=|\frac{3}{2}\frac{-1}{2}\rangle$, $|\Psi_3\rangle=|\frac{3}{2}\frac{1}{2}\rangle$ and $|\Psi_4\rangle=|\frac{3}{2}\frac{3}{2}\rangle$. For this case, we cannot use the same Gell-Mann matrices as we did for spin 1. Therefore, we can use the generalized Gell-Mann matrices, which are a generative group of SU(4)[3]. There are therefore 15 generalized Gell-Mann matrices in this group. The complete set of the generalized Gell-Mann matrices can be found in Appendix B. Using this set of matrices, we can write the spin operator as:

$$S_x=\frac{\sqrt{3}}{2}(\lambda_1+\lambda_{13})+\lambda_6=\frac{1}{2}\begin{pmatrix} 0 & \sqrt{3} & 0 & 0 \\ \sqrt{3} & 0 & 2 & 0 \\ 0 & 2 & 0 & \sqrt{3} \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix}, \tag{19}$$

$$S_y=\frac{\sqrt{3}}{2}(\lambda_2+\lambda_{14})+\lambda_7=\frac{1}{2}\begin{pmatrix} 0 & -i\sqrt{3} & 0 & 0 \\ i\sqrt{3} & 0 & -2i & 0 \\ 0 & 2i & 0 & -i\sqrt{3} \\ 0 & 0 & i\sqrt{3} & 0 \end{pmatrix}, \tag{20}$$

$$S_z=\frac{1}{2}\lambda_3+\frac{1}{\sqrt{3}}\lambda_8+\frac{\sqrt{6}}{3}\lambda_{15}=\begin{pmatrix} 3/2 & 0 & 0 & 0 \\ 0 & 1/2 & 0 & 0 \\ 0 & 0 & -1/2 & 0 \\ 0 & 0 & 0 & -3/2 \end{pmatrix}. \tag{21}$$

Therefore, we can write the Hamiltonian as the following matrix:

$$H=\begin{pmatrix} -\frac{3}{2}B_z+\frac{9}{4}K_\parallel+\frac{3}{4}K_\perp & \frac{-\sqrt{3}}{2}(B_x-iB_y) & \frac{\sqrt{3}}{2}K_\parallel & 0 \\ \frac{-\sqrt{3}}{2}(B_x+iB_y) & \frac{-B_z}{2}+\frac{K_\parallel}{4}+\frac{7K_\perp}{4} & -(B_x-iB_y) & \frac{\sqrt{3}}{2}K_\perp \\ \frac{\sqrt{3}}{2}K_\perp & -(B_x+iB_y) & \frac{1}{2}B_z+\frac{7}{4}K_\perp+\frac{1}{4}K_\parallel & \frac{-\sqrt{3}}{2}(B_x-iB_y) \\ 0 & \frac{\sqrt{3}}{2}K_\perp & \frac{-\sqrt{3}}{2}(B_x+iB_y) & \frac{3}{2}B_z+\frac{3}{4}K_\perp+\frac{9}{4}K_\parallel \end{pmatrix}. \tag{22}$$

Using the spherical coordinates as defined in the previous subsection, we can rewrite this matrix as the following:

$$H = \begin{pmatrix} -\frac{3}{2}B\cos(\theta) + \frac{9}{4}K_{\parallel} + \frac{3}{4}K_{\perp} & \frac{-\sqrt{3}}{2}B\sin(\theta)e^{-i\varphi} & \frac{\sqrt{3}}{2}K_{\parallel} & 0 \\ \frac{-\sqrt{3}}{2}B\sin(\theta)e^{i\varphi} & \frac{-B\cos(\theta)}{2} + \frac{K_{\parallel}}{4} + \frac{7K_{\perp}}{4} & -B\sin(\theta)e^{-i\varphi} & \frac{\sqrt{3}}{2}K_{\perp} \\ \frac{\sqrt{3}}{2}K_{\perp} & -B\sin(\theta)e^{i\varphi} & \frac{1}{2}B\cos(\theta) + \frac{7}{4}K_{\perp} + \frac{1}{4}K_{\parallel} & \frac{-\sqrt{3}}{2}B\sin(\theta)e^{-i\varphi} \\ 0 & \frac{\sqrt{3}}{2}K_{\perp} & \frac{-\sqrt{3}}{2}B\sin(\theta)e^{i\varphi} & \frac{3}{2}B\cos(\theta) + \frac{3}{4}K_{\perp} + \frac{9}{4}K_{\parallel} \end{pmatrix}. \tag{23}$$

We can see here also that the matrix has a pretty complicated form and we will therefore use a Python program to diagonalize it and find the average values of the spin.

2.4 Anti coherent spin states

In order for our simulations to be interesting to look at, we first need to choose the right input states. One of the effects of anisotropy we can expect to see for $s = 1$ and $s = \frac{3}{2}$ is as an example, quantum revival.
Quantum revival is defined by a recurrence of the wave packet. Therefore the pattern of the wave function will repeat itself exactly (full revival), or at different scaled (fractional revival). This effect is mostly seen when the input stat is an anti-coherent state of the spin operator.
This kind of state will have a vanishing $\langle \boldsymbol{S} \rangle$. It is worth noting that most spin states are neither coherent or anti coherent.
A pure spin state can be written as:

$$|\Psi\rangle = \sum_{m=-s}^{s} c_m |s; m\rangle . \tag{24}$$

We also know that we expect the average value of the spin states to be zero, that is:

$$\langle S_z \rangle = \sum_{m=-s}^{s} m|c_m|^2 = 0, \tag{25}$$

$$\langle S_{\pm} \rangle = \sum_{m=-s}^{s-1} c_m^* c_{m+1} \sqrt{s(s+1) - m(m+1)}. \tag{26}$$

The use of the ladder spin operator is most useful as $S_+ = S_x + iS_y$ and $S_- = S_x - iS_y$.
In order for our first condition to be respected, we can parametrize the pure states to have $c_m = e^{i\phi_m} a_{|m|}$ with $\sum_{m=-s}^{s} a_{|m|}^2 = 1$.
For any given $s \geq \frac{1}{2}$, we thus only need to determine the phase $i\Phi$ that makes the ladder spin operators vanish.
The easiest anti coherent state, and the one we will use in our simulations is a Schrödinger cat state [4] defined as:

$$|\Psi_s^{cat}\rangle = \frac{|s; -s\rangle + |s; +s\rangle}{\sqrt{2}}. \tag{27}$$

This corresponds to the states $|\Psi_1^{cat}\rangle = \frac{|1;-1\rangle + |1;+1\rangle}{\sqrt{2}}$ and $|\Psi_{\frac{3}{2}}^{cat}\rangle = \frac{|\frac{3}{2};\frac{-3}{2}\rangle + |\frac{3}{2};\frac{+3}{2}\rangle}{\sqrt{2}}$.

3 Simulations and results

First of all, we can look at the $s = \frac{1}{2}$ case:

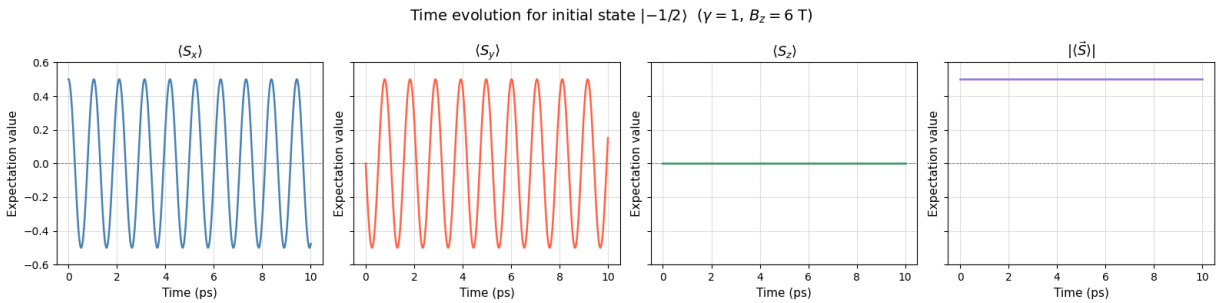
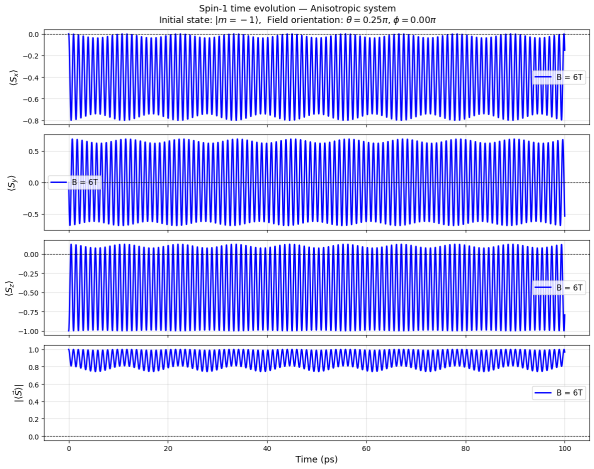
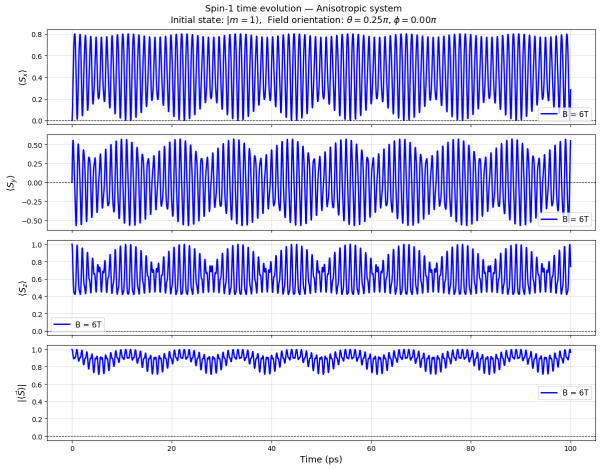


Figure 1: Average values of the spin operator with input state $m = \frac{-1}{2}$

We can see that $\langle S_z \rangle = 0$ as expected and that $\langle S_x \rangle$ and $\langle S_y \rangle$ oscillate. This was the expected result of the simulation.
For the case $s = 1$, the plots for the initial states $m = -1$ and $m = 1$ look very similar and are not very interesting to analyze as there is no clear oscillation pattern. This can be seen with the plot below.



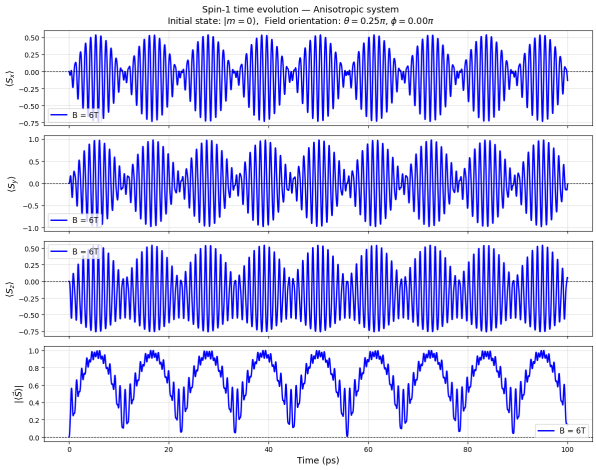
(a) Average values of the spin operator with input state $m = -1$



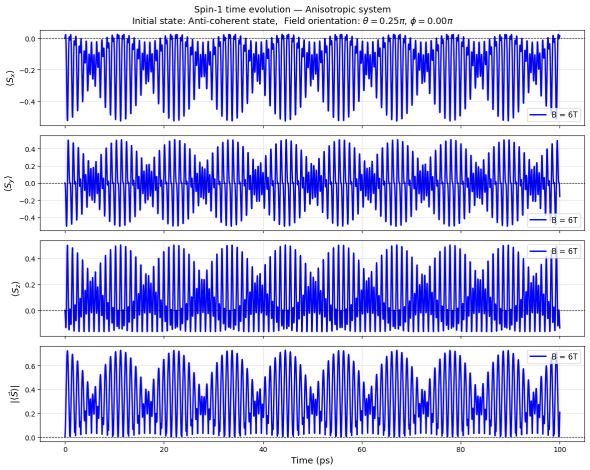
(b) Average values of the spin operator with input state $m = 1$

Figure 2: Side by side comparison of the average values of the spin operator for the input states $m = 1$ and $m = -1$

However, the plots for the input state $m = 0$ and the anti-coherent state show a repeating pattern of oscillations. This can be seen in the two figures below.



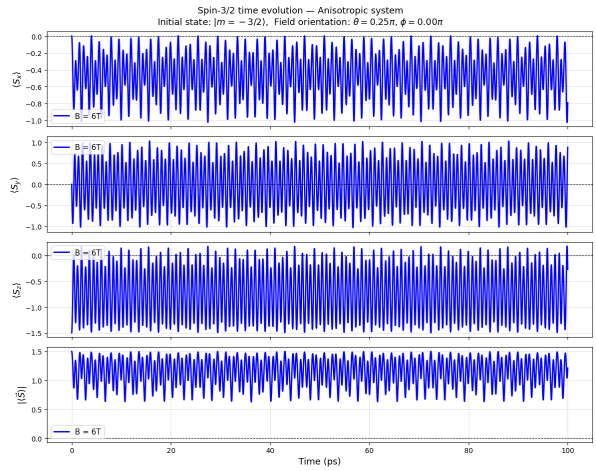
(a) Average values of the spin operator with input state $m = 0$



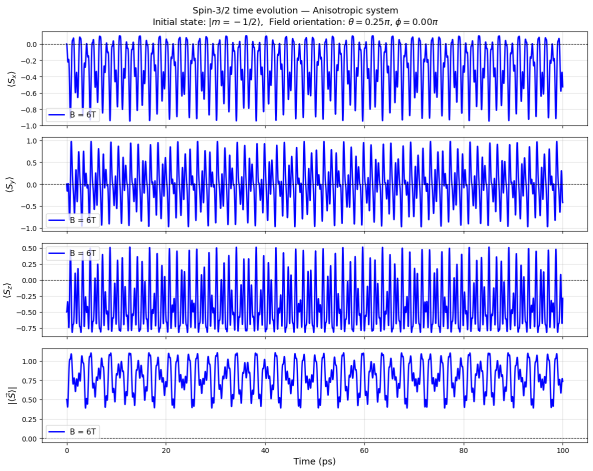
(b) Average values of the spin operator with anti coherent input state

Figure 3: Side by side comparison of the average values of the spin operator for the anti coherent input state and the $m = 0$ input state

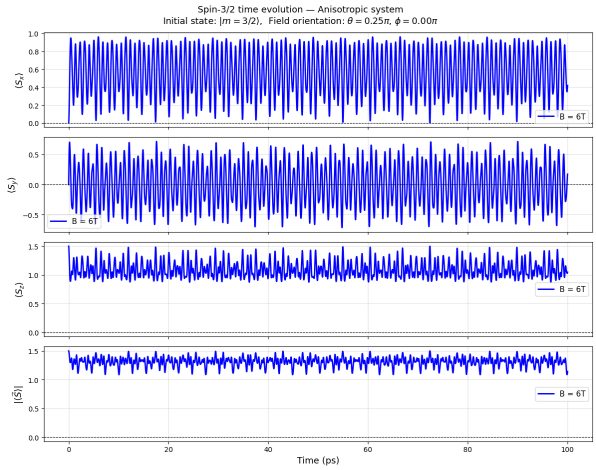
We can see in those figures an oscillation pattern. This is due to quantum revival, a phenomena described in the precedent section. What we can observe here seems to be a full revival of the wave packet. We can indeed observe that the state $m = 0$ is also an anti coherent state despite not being a Schrödinger cat state.
In the case $s = \frac{3}{2}$, no clear oscillation pattern is seen, even with an anti coherent state. This might be caused by the phase factors when computing the system. For a full quantum revival to happen, all the phases must return to 1 at the same time. Partial revivals will then occur at fractions of the full revival time. When no clear oscillation pattern is seen, it means that there are interferences between the different phases. This can be seen by the states $m = 1$ and $m = -1$ for spin 1, but also for the cases $m = \frac{-3}{2}, \frac{-1}{2}, \frac{1}{2}, \frac{3}{2}$ and the Schrödinger cat state. This can be seen in the plots below.



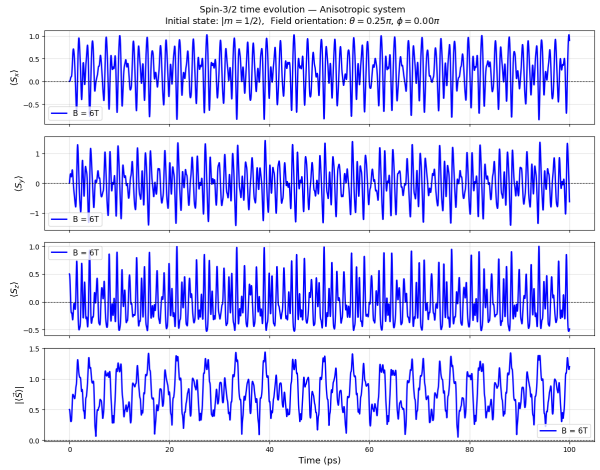
(a) Average values of the spin operator with input state $m = \frac{-3}{2}$



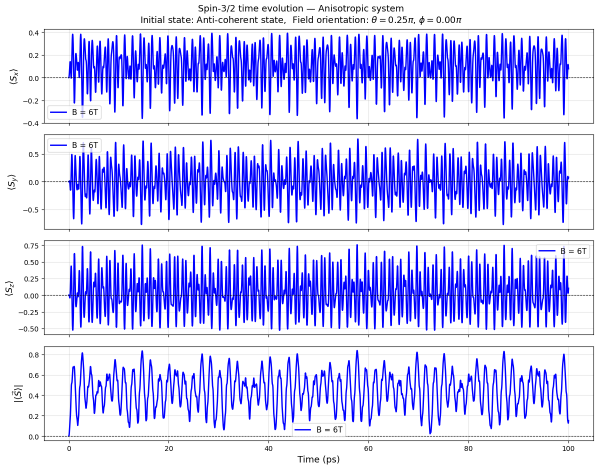
(b) Average values of the spin operator with input state $m = \frac{-1}{2}$



(a) Average values of the spin operator with input state $m = \frac{3}{2}$



(b) Average values of the spin operator with input state $m = \frac{1}{2}$



(c) Average values of the spin operator with anti coherent input state

Figure 5: Average values of the spin for different input states of $s = \frac{3}{2}$

4 Conclusion

In this small project, we have studied the spin dynamics for three different spins ($s = \frac{1}{2}$, $s = 1$ and $s = \frac{3}{2}$) for a system in the presence of anisotropy. We have thus been able to show the presence of full quantum revival for the $m = 0$ input state and a Schrödinger cat input state.
The understanding of higher spin dynamics in the presence of anisotropy is extremely important, as it enables better control when manipulating qubits and qudits.

References

[1] Earl T. Campbell. Enhanced fault-tolerant quantum computing in d-level systems. ISSN 0031-9007, 1079-7114. URL http://arxiv.org/abs/1406.3055.

[2] Simplifying quantum logic using higher-dimensional hilbert spaces. URL https://www.researchgate.net/publication/43512561_Simplifying_quantum_logic_using_higher-dimensional_Hilbert_spaces.

[3] Reinhold A. Bertlmann and Philipp Krammer. Bloch vectors for qudits. URL http://arxiv.org/abs/0806.1174.

[4] D. Baguette and J. Martin. Anticoherence measures for pure spin states. 96.

A Gell-Mann matrices

The full set of Gell-Mann matrices can be found here :

$$\lambda_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \qquad \lambda_2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \qquad \lambda_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\lambda_4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \qquad \lambda_5 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix} \qquad \lambda_6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\lambda_7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix} \qquad \lambda_8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}$$

B Generalized Gell-Mann matrices

The full set of genralized matrices for SU(4) can be found here :

$$\lambda_1 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \lambda_2 = \begin{pmatrix} 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \lambda_3 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\lambda_4 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \lambda_5 = \begin{pmatrix} 0 & 0 & -i & 0 \\ 0 & 0 & 0 & 0 \\ i & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \lambda_6 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\lambda_7 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -i & 0 \\ 0 & i & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \lambda_8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \lambda_9 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

$$\lambda_{10} = \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix}, \qquad \lambda_{11} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \qquad \lambda_{12} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -i \\ 0 & 0 & 0 & 0 \\ 0 & i & 0 & 0 \end{pmatrix}$$

$$\lambda_{13} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \lambda_{14} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \end{pmatrix}, \qquad \lambda_{15} = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -3 \end{pmatrix}$$